

Computational Validation of Adaptive Interface Control in TMI Quantum Systems

Extended Technical Report on Computational Tests 1–11

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May 6, 2026

Abstract

This document consolidates the computational test line for quantum-system predictions in the Theory of Manifold Interfaces (TMI). This version extends the initial Tests 1–6 and adds Tests 7–11: full density-matrix Monte Carlo validation, cross-model validation, failure-boundary mapping, adversarial failure testing, and safety-aware recovery testing.

The tested engineering core has the form

$$I_Q^{coh-auto} \Rightarrow Loss_{coh} \Downarrow \Rightarrow T_2^{eff} \Uparrow \Rightarrow P_{success} \Uparrow.$$

The report does not claim a laboratory proof of the physical existence of the field Ξ_I^{auto} . It fixes a more modest result: in computational models, adaptive interface control can increase effective coherence and circuit success probability, but it has a domain of applicability and requires a safety-aware layer under adversarial conditions.

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1 Status and Scientific Caution

This document is a **computational validation report**, not a laboratory confirmation. We do not attempt to verify the entire Theory of Manifold Interfaces at once. We test one narrow consequence:

$$\boxed{I_Q^{coh-auto} \Rightarrow \gamma_{eff} \Downarrow \Rightarrow T_2^{eff} \Uparrow \Rightarrow P_{success} \Uparrow}$$

where $I_Q^{coh-auto}$ is a coherence-preserving automatic interface, $\gamma_{eff} = 1/T_2^{eff}$ is the effective decoherence rate, and $P_{success}$ is the probability of obtaining the correct result in a small quantum circuit.

Starting with Test 10 and Test 11, the report also fixes a negative and safety-related consequence:

$$\boxed{\text{Auto}(I) \not\Rightarrow \text{Safe}(I)}$$

and

$$\boxed{\text{Auto}(I) + \text{SafetyLayer} \Rightarrow \text{SaferAdaptiveControl.}}$$

In other words, adaptivity by itself is not a guarantee of safety: under misleading feedback and high control cost, an automatic interface can lose to a fixed standard control mode.

2 Connection to TMI Core 0.2 and TMI-Physics 2.2

The computational test line does not replace Core 0.2 or TMI-Physics 2.2. Its correct role is that of an applied computational module:

$$\boxed{\text{TMI Core 0.2} \Rightarrow \text{TMI-Physics 2.2} \Rightarrow \text{Computational Validation}}$$

In Core 0.2, automaticity is understood as a definitional property of a valid interface:

$$\text{Valid}(I) \Rightarrow \text{Auto}(I), \quad \text{Auto}(I) \not\Rightarrow \text{AlwaysAct}(I).$$

In TMI-Physics 2.2 this logic is projected into the physical layer:

$$\Xi_I \in \mathfrak{I}_{phys} \Rightarrow \text{Auto}(\Xi_I) \Rightarrow \Xi_I^{auto}.$$

The computational tests examine an engineering projection of this principle: can an automatic coherence-control interface improve observable metrics in quantum-system models?

3 Summary of Tests 1–11

For Tests 1–4, the final column typically lists the values in the order base / standard / TMI.

Test	Meaning	Criterion	Result
Test 1	Ramsey toy model	$T_2^{TMI} > T_2^{std} > T_2^{base}$	50.0 / 120.0 / 178.6 μs
Test 2	Noisy Ramsey learning	$T_2^{TMI} > T_2^{std} > T_2^{base}$	51.3 / 125.8 / 162.4 μs
Test 3	Small circuit	$P_{TMI} > P_{std} > P_{base}$	0.182 / 0.386 / 0.431

Test	Meaning	Criterion	Result
Test 4	Stabilized interface	$P_{stab} > P_{std} > P_{base}$	0.180 / 0.328 / 0.465
Test 5	Surrogate Monte Carlo	$E[P_{TMI}] > E[P_{std}]$	mean $\Delta = 0.0512$, $Pr = 0.864$
Test 6	Ablation study	full TMI > standard	full $\Delta \approx 0.0677$
Test 7	Full density-matrix MC	$E[P_{TMI}] > E[P_{std}]$	mean $\Delta = 0.0935$, $Pr = 0.933$
Test 8	Cross-model validation	advantage across circuit families	mean $\Delta = 0.0662$, $Pr = 0.846$
Test 9	Failure-boundary map	locate weak/failure regions	failure cells = 0, weak cells = 6
Test 10	Adversarial failure	force $P_{TMI} \leq P_{std}$	failure cells = 58, weak cells = 22
Test 11	Safety recovery	reduce failure rate	$\Delta_{safety} = 0.0139$, recovered cells = 31/58

4 Tests 1–6: Basic Computational Validation Line

4.1 Test 1: Ramsey toy model

The coherence model is

$$C(t) = C_0 e^{-t/T_2}.$$

The result is

$$T_2^{base} = 50.0 \mu s, \quad T_2^{std} = 120.0 \mu s, \quad T_2^{TMI} = 178.6 \mu s.$$

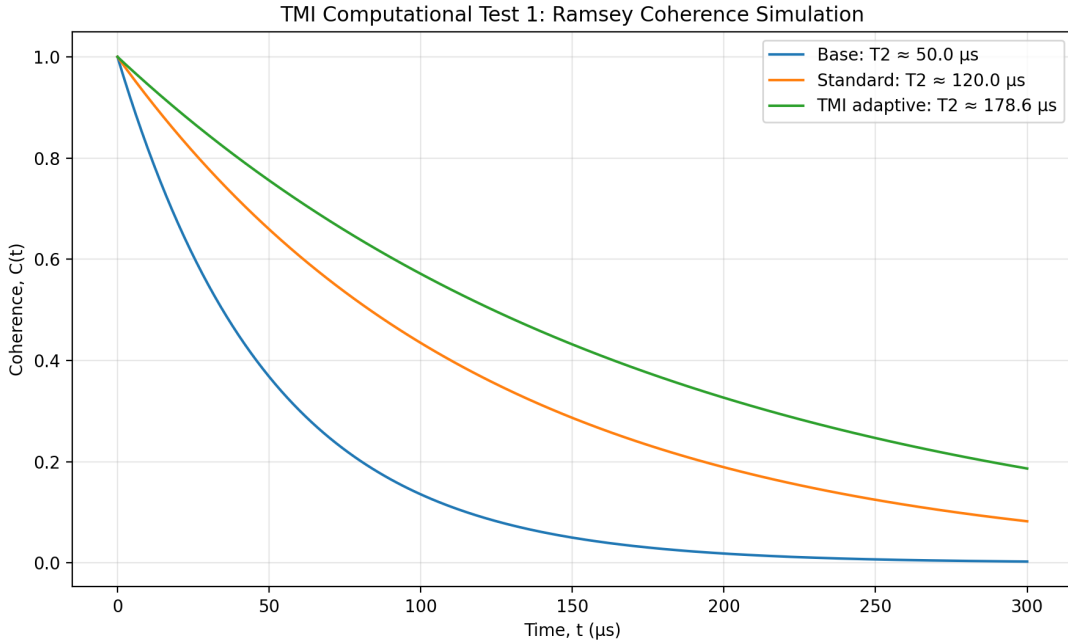


Figure 1: Test 1: simple Ramsey coherence model.

4.2 Test 2: Noisy Ramsey learning

The automatic interface must learn the protection parameter u from noisy Ramsey observations. The result is

$$T_2^{base} = 51.3 \mu s, \quad T_2^{std} = 125.8 \mu s, \quad T_2^{TMI} = 162.4 \mu s.$$

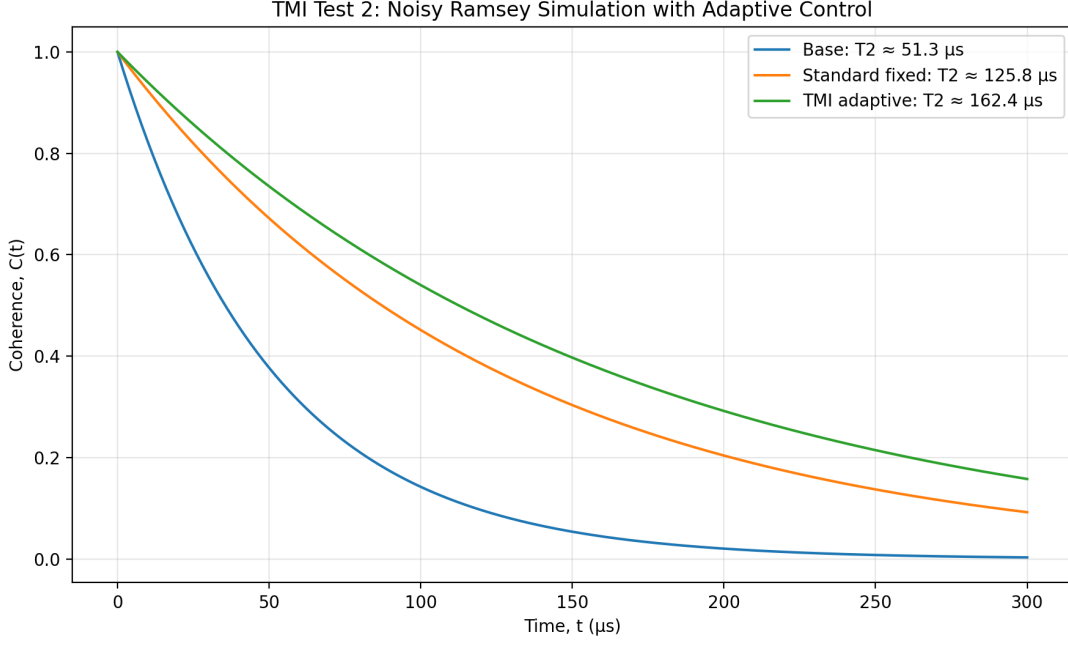


Figure 2: Test 2: Ramsey curves under noisy observations.

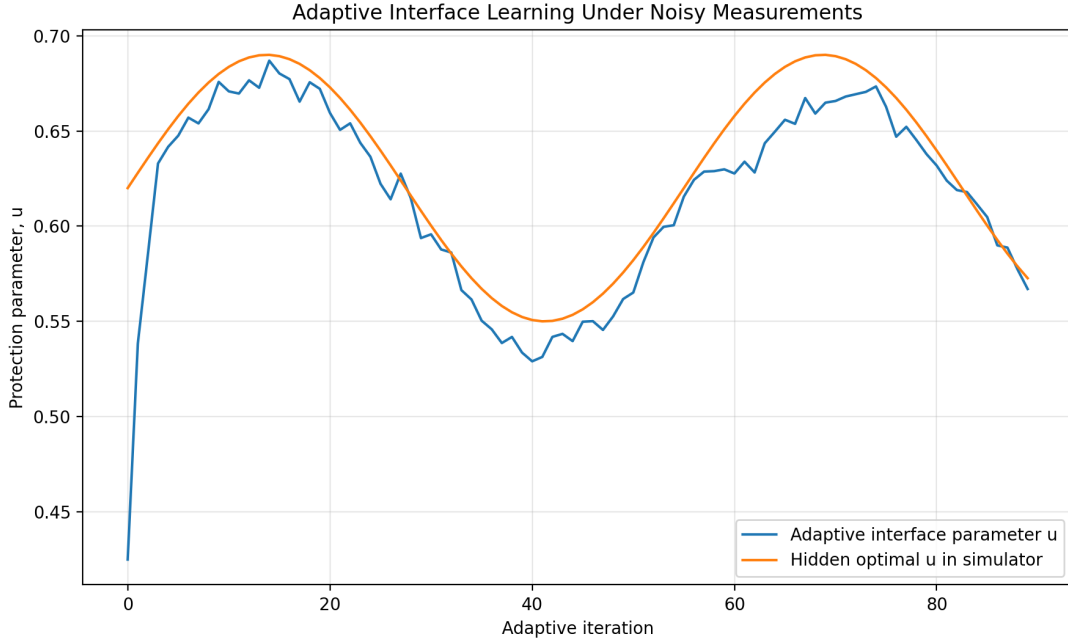


Figure 3: Test 2: learning of the protection parameter u .

4.3 Test 3: Small quantum circuit

The circuit is constructed as

$$|000\rangle \rightarrow \text{entangling circuit} \rightarrow \text{inverse circuit} \rightarrow |000\rangle.$$

The ideal success probability is

$$P_{\text{success}} = \Pr(000).$$

The result is

$$P_{\text{base}} = 0.182, \quad P_{\text{std}} = 0.386, \quad P_{\text{TMI}} = 0.431.$$

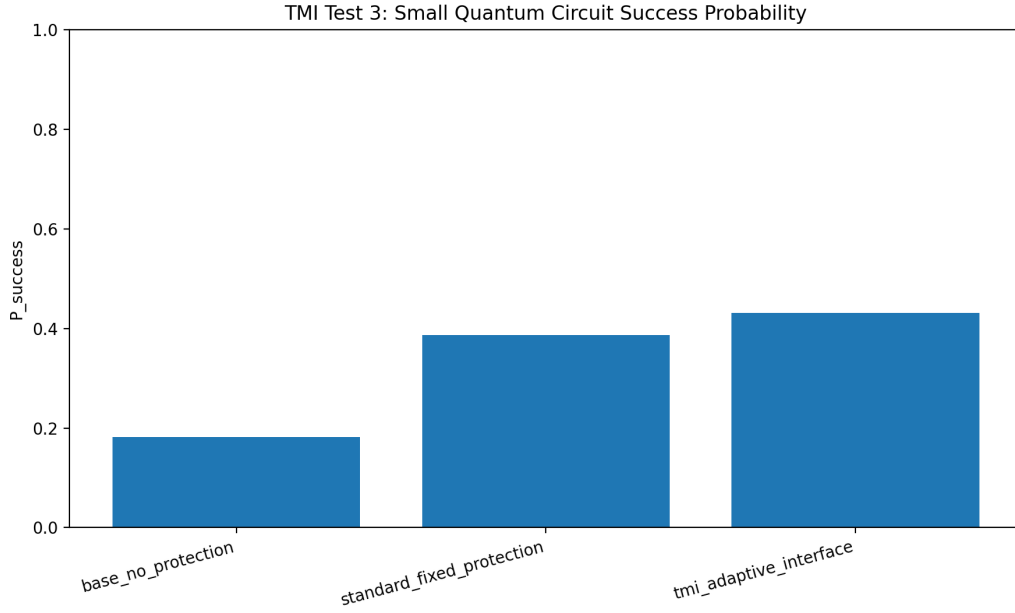


Figure 4: Test 3: success probability of a small quantum circuit.

4.4 Test 4: Stabilized adaptive interface

Test 4 adds memory, smoothing, step limitation, and repeated observations:

$$\text{AutoInterface} + \text{Memory} + \text{Smoothing} + \text{StepLimit} \Rightarrow \text{StableControl}.$$

The result is

$$P_{\text{base}} = 0.180, \quad P_{\text{std}} = 0.328, \quad P_{\text{raw}} = 0.465, \quad P_{\text{stab}} = 0.465.$$

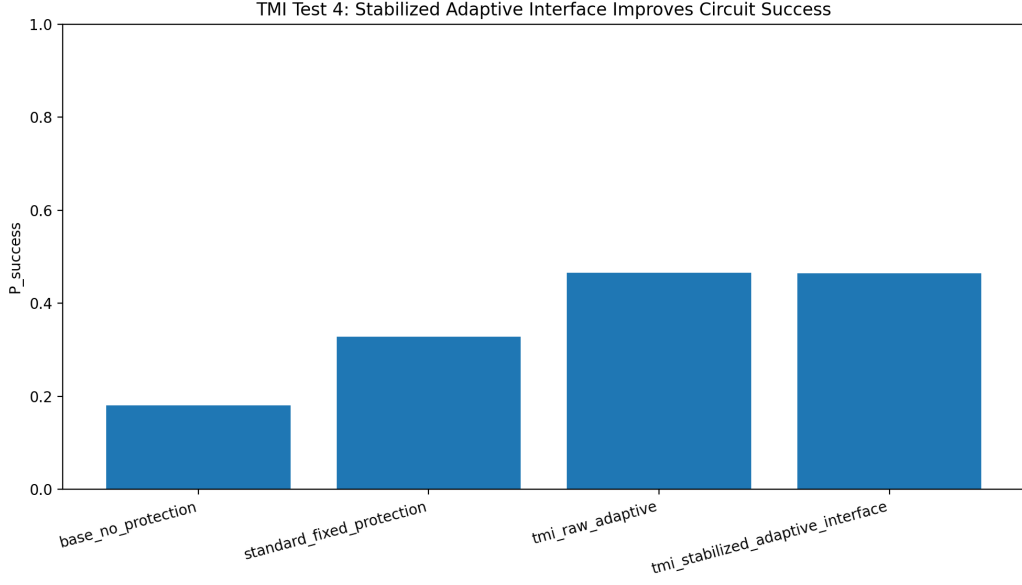


Figure 5: Test 4: stabilized adaptive interface improves $P_{success}$.

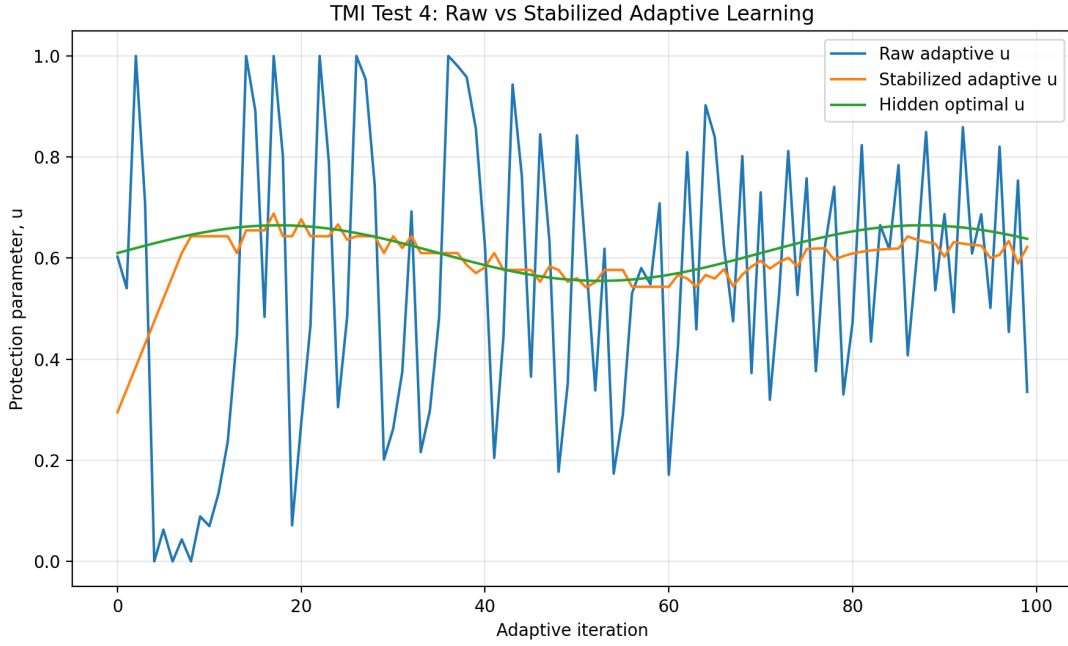


Figure 6: Test 4: raw adaptive control jumps, while stabilized adaptive control keeps the parameter closer to the optimal regime.

4.5 Test 5: Monte Carlo surrogate validation

Test 5 checks statistical advantage across 500 random environments using a fast circuit-level surrogate:

$$\mathbb{E}[P_{TMI}] > \mathbb{E}[P_{std}], \quad \Pr(P_{TMI} > P_{std}) > 0.5.$$

The result is

$$\begin{aligned} \mathbb{E}[P_{base}] &= 0.245, & \mathbb{E}[P_{std}] &= 0.417, & \mathbb{E}[P_{TMI}] &= 0.468. \\ \mathbb{E}[P_{TMI} - P_{std}] &= 0.0512, & \Pr(P_{TMI} > P_{std}) &= 0.864. \end{aligned}$$

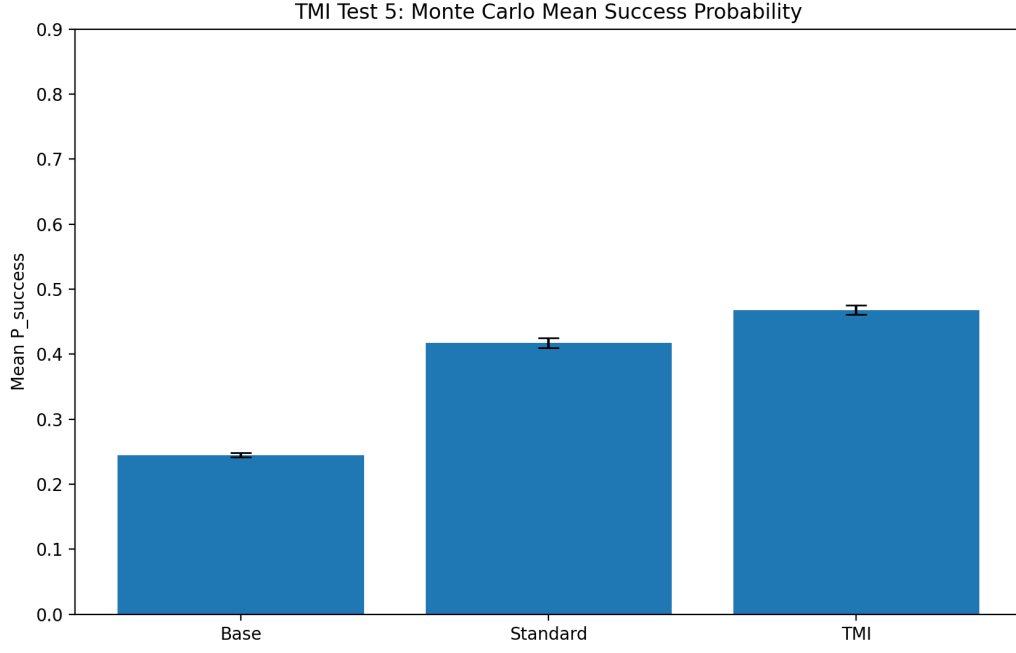


Figure 7: Test 5: mean $P_{success}$ in Monte Carlo surrogate validation.

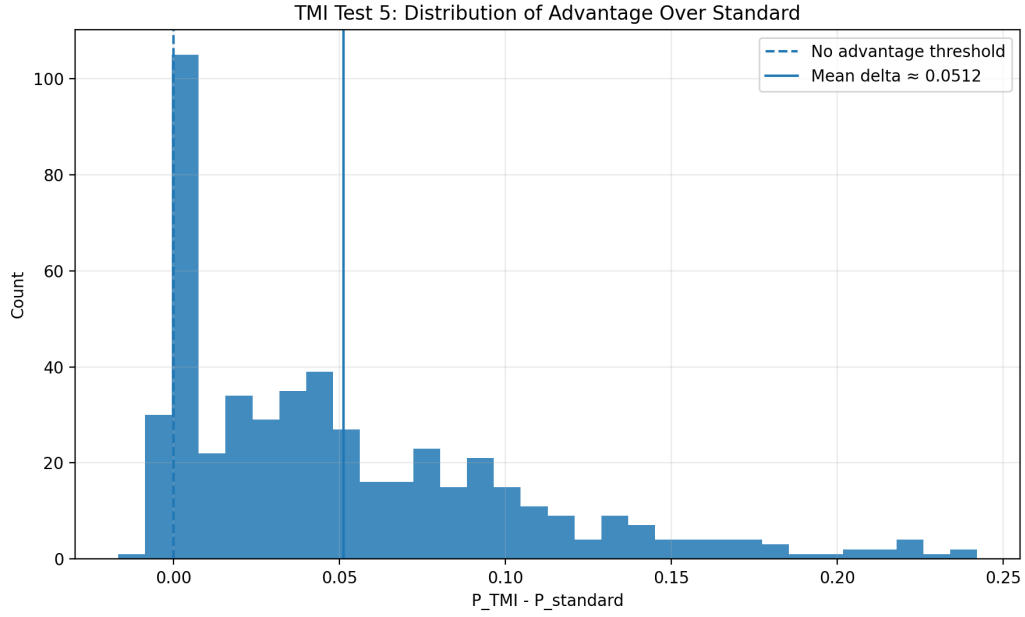


Figure 8: Test 5: distribution of the advantage $P_{TMI} - P_{std}$.

4.6 Test 6: Ablation study

Test 6 shows that raw adaptivity alone does not guarantee an improvement:

$$\text{RawAdaptivity} \not\Rightarrow \text{Improvement}.$$

The advantage appears as a property of a structured automatic interface:

$$\text{StructuredAutoInterface} \Rightarrow \text{StableParameterTracking} \Rightarrow P_{success} \uparrow.$$

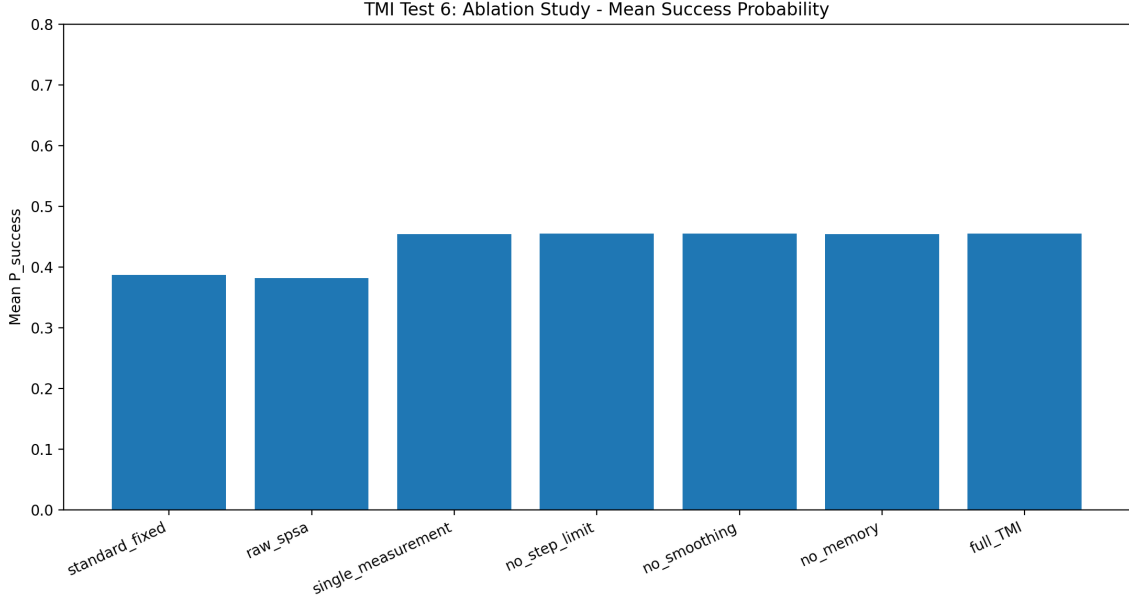


Figure 9: Test 6: ablation study across interface-control configurations.

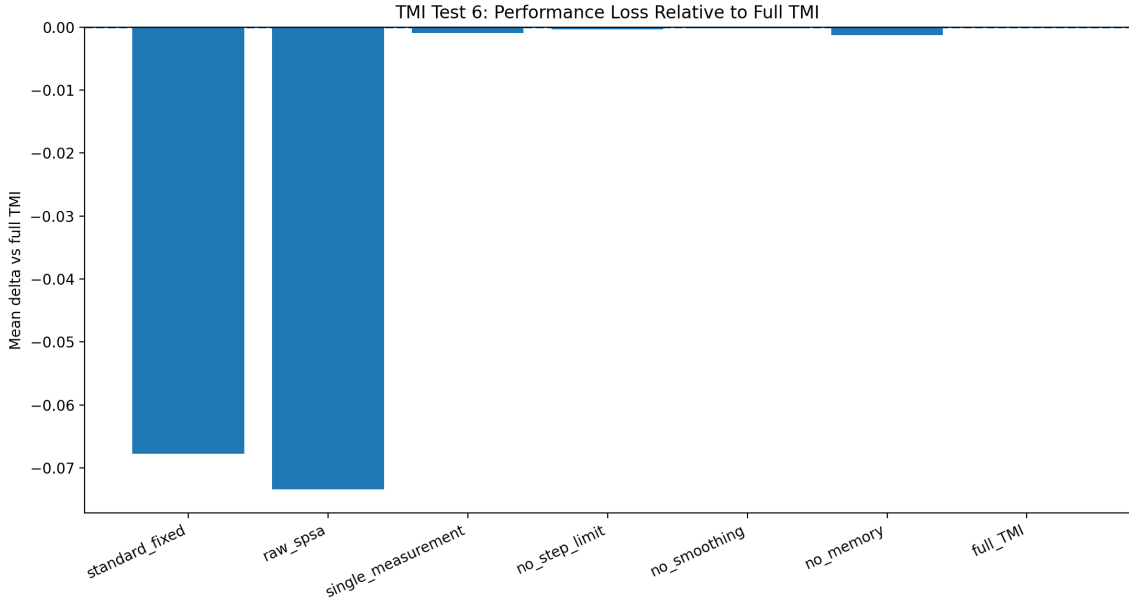


Figure 10: Test 6: losses relative to the full TMI mechanism.

5 Tests 7–8: Robustness and Cross-Model Validation

5.1 Test 7: Full density-matrix Monte Carlo

Test 7 repeats the statistical validation of Test 5, but now on the full 3-qubit density-matrix toy model. This removes the main limitation of Test 5: dependence on a fast surrogate.

$$\begin{aligned}\mathbb{E}[P_{\text{base}}] &= 0.182, & \mathbb{E}[P_{\text{std}}] &= 0.322, & \mathbb{E}[P_{\text{TMI}}] &= 0.415. \\ \mathbb{E}[P_{\text{TMI}} - P_{\text{std}}] &= 0.0935, & \Pr(P_{\text{TMI}} > P_{\text{std}}) &= 0.933.\end{aligned}$$

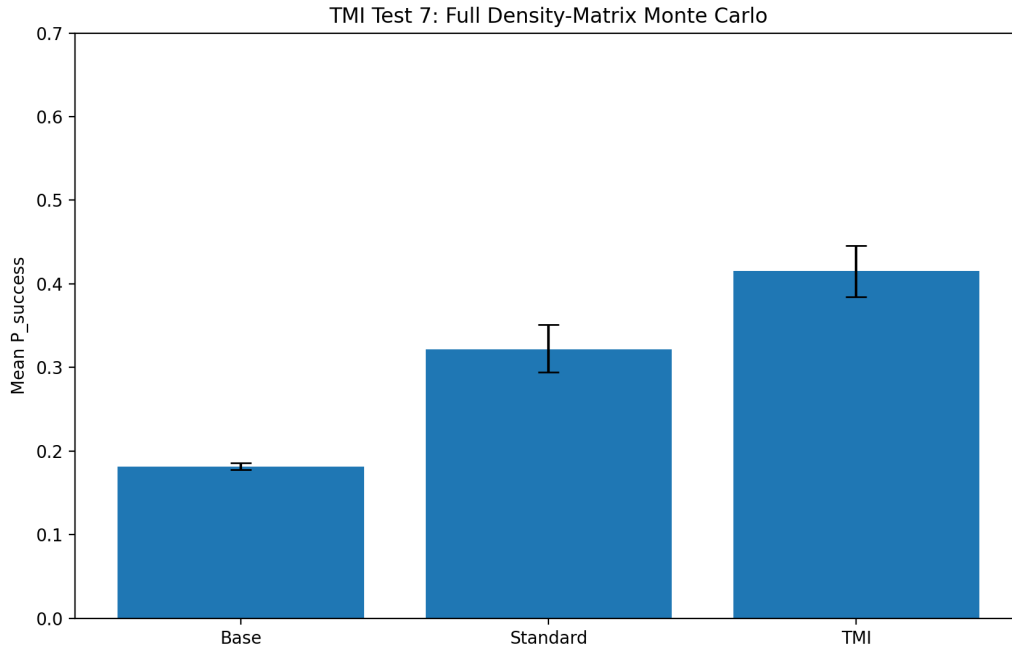


Figure 11: Test 7: full density-matrix Monte Carlo, mean P_{success} .

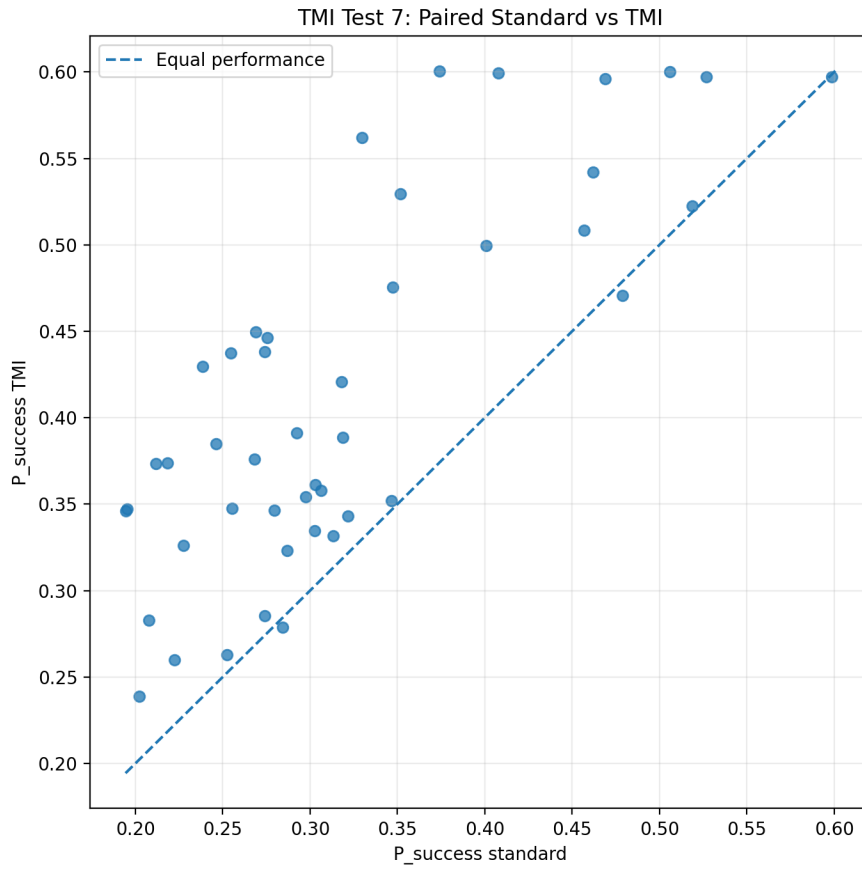


Figure 12: Test 7: paired comparison P_{std} vs P_{TMI} .

5.2 Test 8: Cross-model validation

Test 8 checks not one circuit, but four circuit families. The overall result is

$$\mathbb{E}[P_{base}] = 0.292, \quad \mathbb{E}[P_{std}] = 0.379, \quad \mathbb{E}[P_{TMI}] = 0.445.$$

$$\mathbb{E}[P_{TMI} - P_{std}] = 0.0662, \quad \Pr(P_{TMI} > P_{std}) = 0.846.$$

Circuit	Base	Standard	TMI	Δ	Pr
balanced_entangle_inverse	0.169	0.277	0.376	0.0994	1.000
deep_chain_inverse	0.253	0.322	0.377	0.0553	0.923
ghz_long_idle_inverse	0.495	0.512	0.527	0.0148	0.462
local_mix_inverse	0.249	0.404	0.500	0.0955	1.000

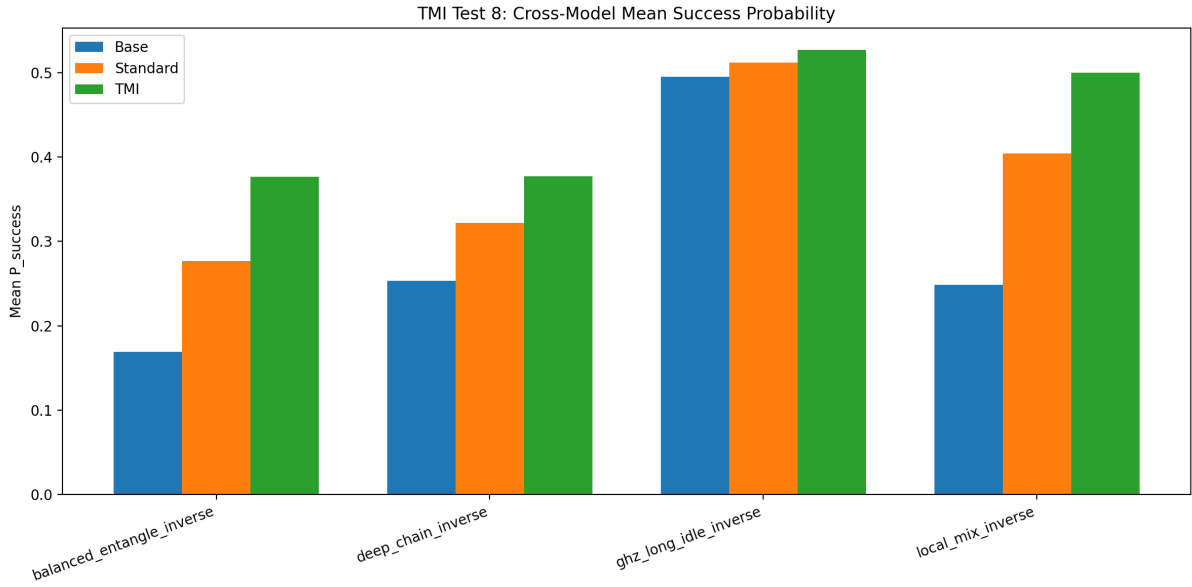


Figure 13: Test 8: mean $P_{success}$ by circuit family.

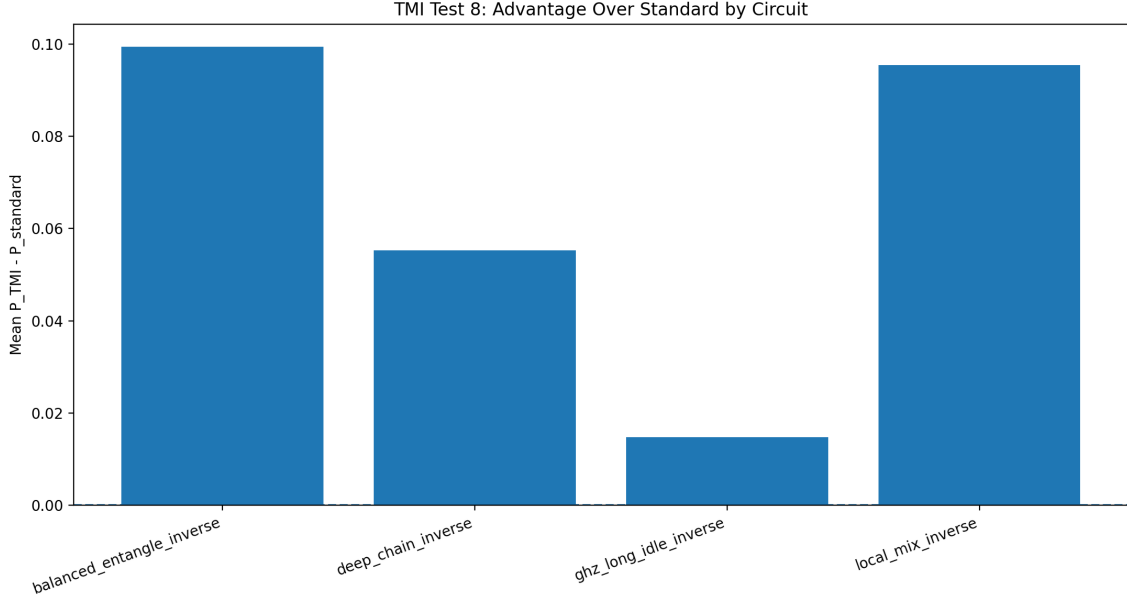


Figure 14: Test 8: TMI advantage over standard control by circuit family.

Important nuance. For the `ghz_long_idle_inverse` circuit, the mean advantage is positive but weaker and less stable than for the other circuit families. This shows that the mechanism is not equally strong for all circuit types.

6 Tests 9–11: Applicability Boundaries and Safety Recovery

6.1 Test 9: Failure-boundary test

Test 9 searches not only for success, but also for weakening zones. We vary the drift speed of the hidden optimum and the measurement noise. The failure criterion is

$$\mathbb{E}[P_{TMI} - P_{std}] \leq 0 \quad \text{or} \quad \Pr(P_{TMI} > P_{std}) \leq 0.5.$$

The overall result is

$$\mathbb{E}[P_{TMI} - P_{std}] = 0.0606, \quad \Pr(P_{TMI} > P_{std}) = 0.849.$$

Strict failure cells found:

0,

weak zones found:

6.

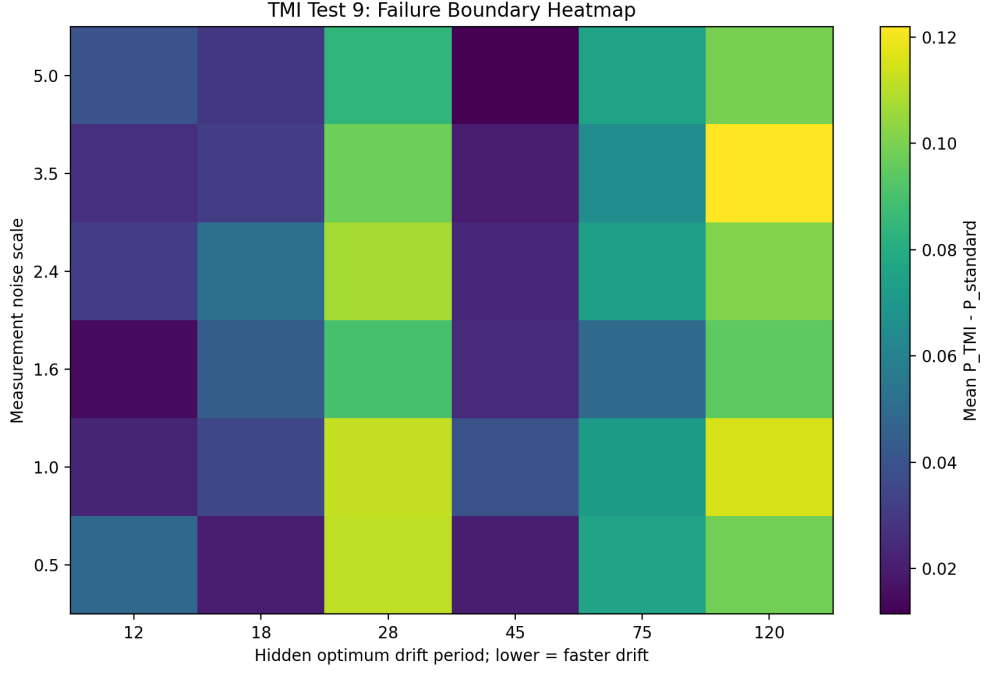


Figure 15: Test 9: heatmap of the advantage $P_{TMI} - P_{std}$ over drift/noise.

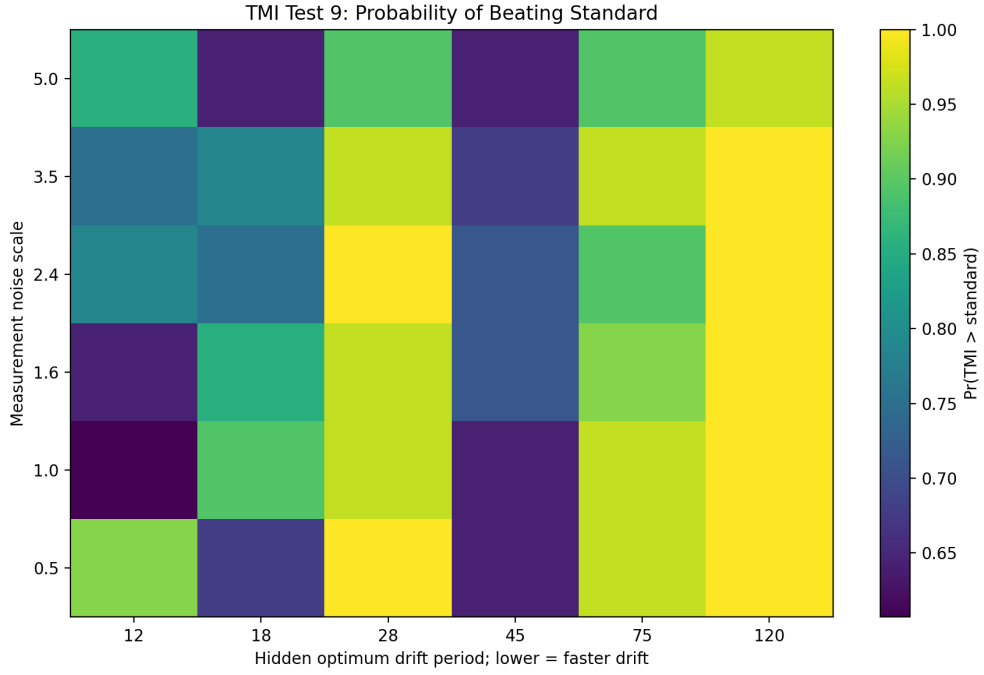


Figure 16: Test 9: probability $\Pr(P_{TMI} > P_{std})$ over drift/noise.

6.2 Test 10: Adversarial failure test

Test 10 intentionally creates adversarial conditions:

fast drift + high noise + misleading feedback + high control penalty.

The goal is to obtain real cases of

$$P_{TMI} \leq P_{std}.$$

The result is

$$\mathbb{E}[P_{TMI} - P_{std}] = 0.0025, \quad \Pr(P_{TMI} > P_{std}) = 0.530.$$

Failure cells:

58,

weak cells:

22.

The key conclusion is

AdaptiveInterface $\not\Rightarrow$ SafeInterface.

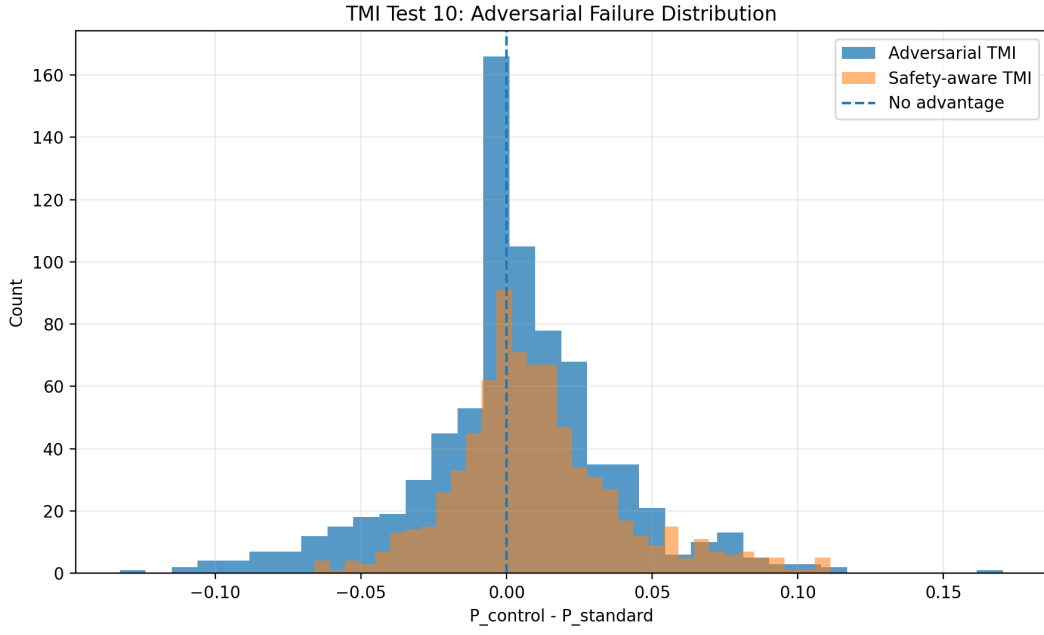


Figure 17: Test 10: adversarial failure distribution.

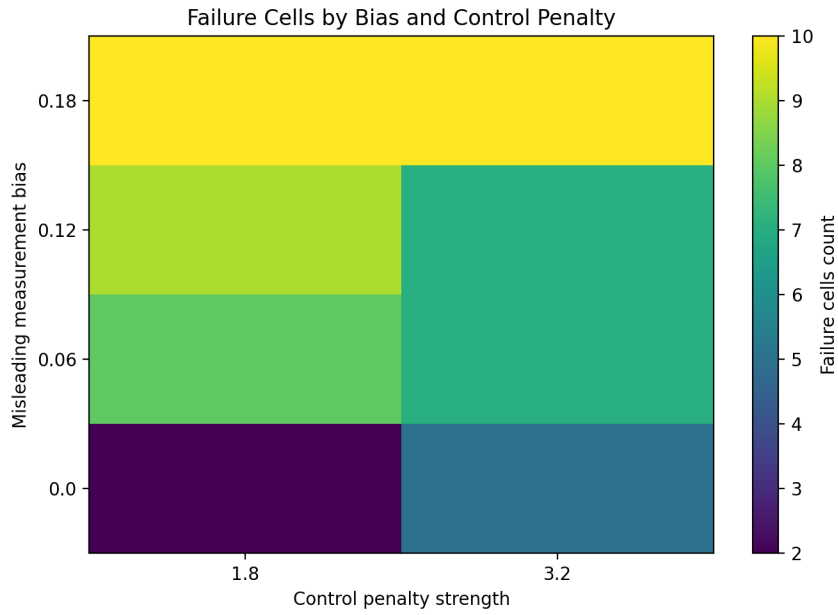


Figure 18: Test 10: number of failure cells by misleading bias and control penalty.

6.3 Test 11: Safety-aware recovery

Test 11 takes only the failure cells from Test 10 and checks whether the advantage can be recovered by a safety layer:

AutoInterface + SafetyPenalty + ConservativeFallback + ValidationChannel $\Rightarrow$ SaferControl.

Mode	Mean P	Δ vs std	$Pr(> std)$	Failure rate	Mean u
standard	0.2799	0.00000	0.000	–	0.450
adversarial_TMI	0.2799	-0.00002	0.478	0.522	0.472
safety_v1	0.2939	0.01392	0.617	0.383	0.395
recovery_v2	0.2932	0.01328	0.579	0.421	0.386

The main result is

$$\Delta_{adv} = -0.00002, \quad \Delta_{safety} = 0.01392, \quad \Delta_{recovery} = 0.01328.$$

Recovery v2 recovered

31/58

failure cells.

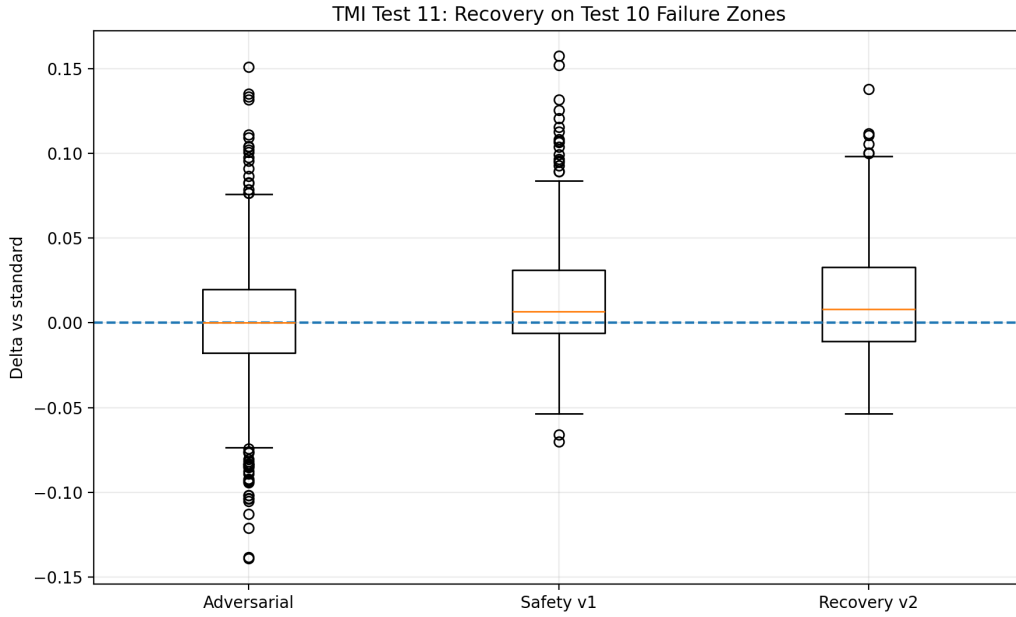


Figure 19: Test 11: safety recovery on failure zones from Test 10.

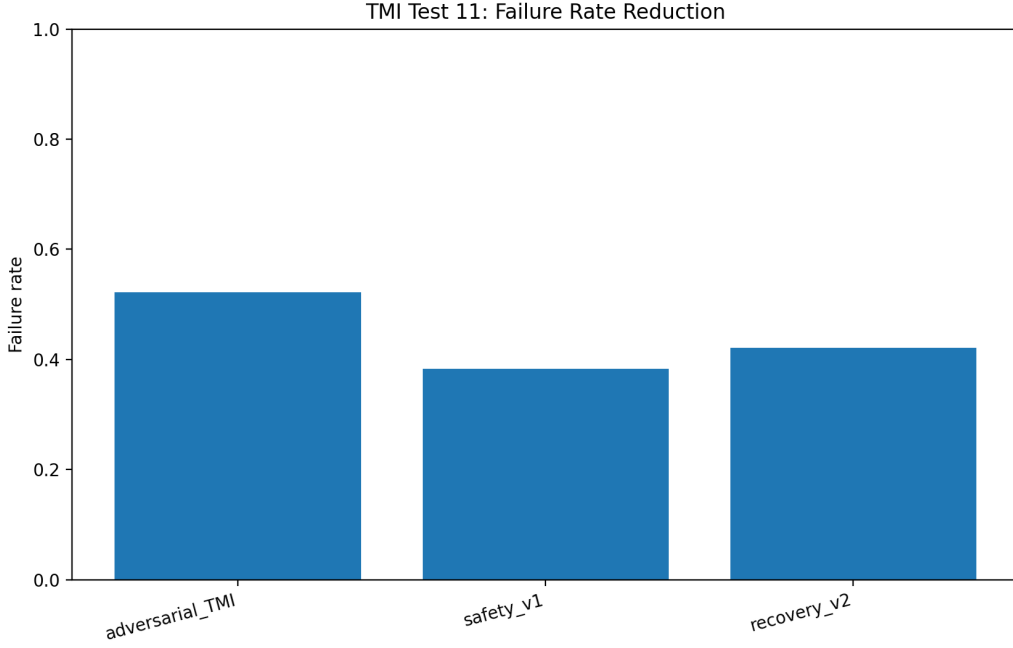


Figure 20: Test 11: failure-rate reduction under safety-aware control.

7 Final Theoretical Consequences

7.1 Positive consequence

Under ordinary and moderately noisy conditions, the computational line supports the engineering hypothesis

$$I_Q^{coh-auto} \Rightarrow T_2^{eff} \uparrow \Rightarrow P_{success} \uparrow$$

7.2 Negative consequence

Test 10 introduces an important boundary:

$$\text{MisleadingFeedback} \wedge \text{HighControlPenalty} \Rightarrow P_{TMI} \leq P_{std}$$

Thus, an automatic interface can make errors if its feedback channel systematically misleads it.

7.3 Safety-aware consequence

Test 11 shows that a safety layer reduces the failure region, but does not eliminate it completely:

$$\text{AutoInterface} + \text{SafetyLayer} \Rightarrow \text{SaferAdaptiveControl}$$

but not

$$\text{AutoInterface} + \text{SafetyLayer} \Rightarrow \text{PerfectControl}.$$

8 Next Steps

The next steps are:

1. Prepare an independent implementation of Tests 7–8 in a Qiskit/Cirq-style framework.
2. Repeat the key tests on a standard noise simulator.
3. Separate the theoretical TMI-Physics 2.2 module from the computational supplement.
4. Prepare a reproducible repository structure: code, data, figures, and reports.
5. After independent simulator validation, move to a cloud-QPU protocol.

9 Conclusion

Tests 1–8 show that adaptive interface control can improve T_2^{eff} and $P_{success}$ in numerical models. Tests 9–11 make the line scientifically more honest: they show not only the region of success, but also the region of weakness, failure, and partial recovery through a safety-aware layer.

The final formula of the report is

$$\boxed{\text{Adaptive Interface Control} \Rightarrow \text{Measurable Computational Advantage}}$$

provided that the environment is observable enough, the feedback channel is stable enough, and control is constrained by safety-aware limits.

A Package Contents

The report archive contains:

- the PDF and LaTeX source of this report;
- figures/ - key plots from Tests 1–11;
- data/ - CSV results and summary files;
- test_archives/ - archives of individual Tests 1–11.